Introduction

Instructor: Mark Kramer

Read the syllabus

Link

EN.553.696:	**Methods in Computational Neuroscience (Fall 2026)**
Instructor:	Mark Kramer (mkramer@jhu.edu)
Course Hours:	Tuesday & Thursday, 9:00-10:15 AM, Maryland 109
Office Hours:	TBD
Textbook:	None
Course Website:	https://mark-kramer.github.io/JHU-553-696
Prerequisites:	Calculus (Differential Equations, Linear Algebra), Computer Science (Programming) or consent of Instructor.

AI use is ok

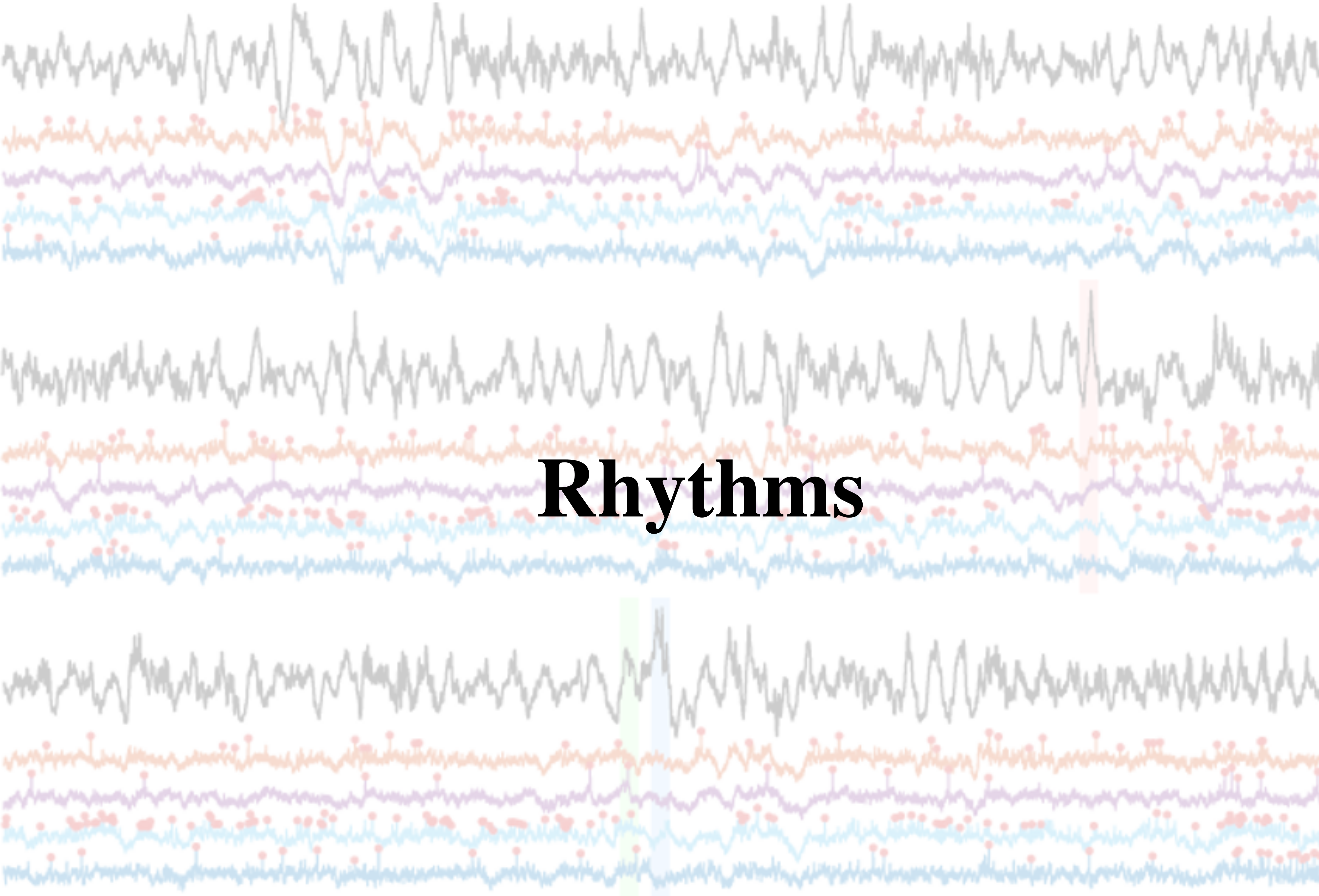
… but follow this rule-of-thumb

if the authors cannot convincingly demonstrate that they are able to give a clear, expert-level talk on their results, one that is correct and properly attributed, then the result should not be published.

[Tao, Mathematics in the Age of AI, arXiv, 2026]

Could I clearly and thoroughly explain every step of this solution to a classmate, without relying on AI?

So, explain your homework during office hours, explain your skill demonstrations to the class.



Rhythms

Brain rhythms

Introduction

What are they?

Where do they come from?

What do they do?

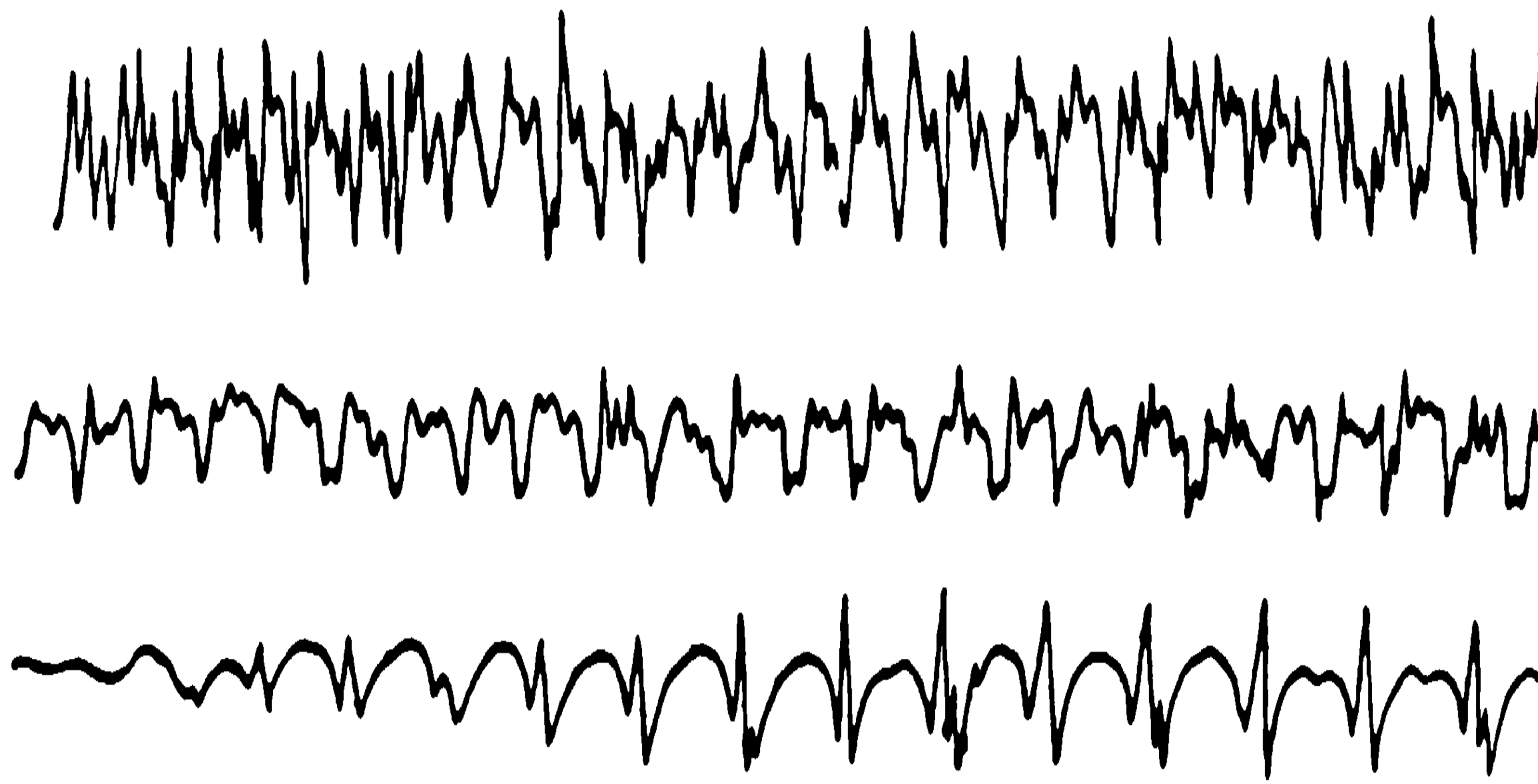
One method of analysis

Brain rhythms

<u>Fact</u>: The brain can generate <u>rhythmic activity</u>.

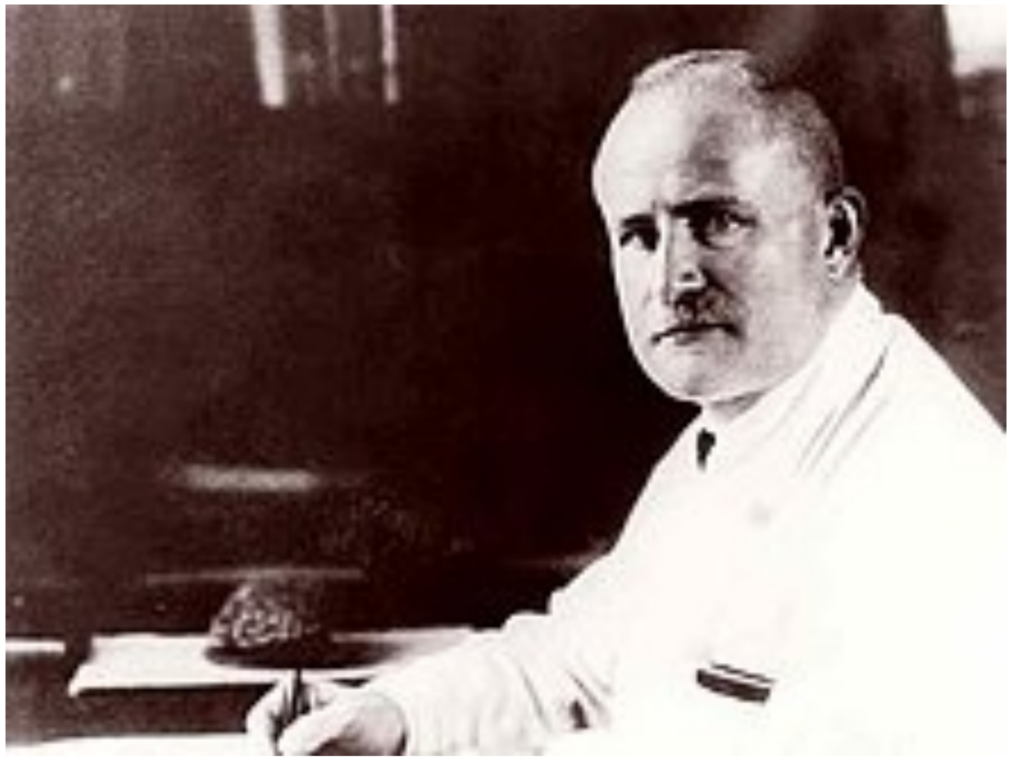
Ex: Scalp electroencephalogram (EEG)

<u>Note</u>: Rhythms also appear in LFP, MEG, fMRI, ...



<u>Observe</u>: Different shapes. Different frequencies.

Brain rhythms



Hans Berger (1873-1941)

Inventor of the EEG

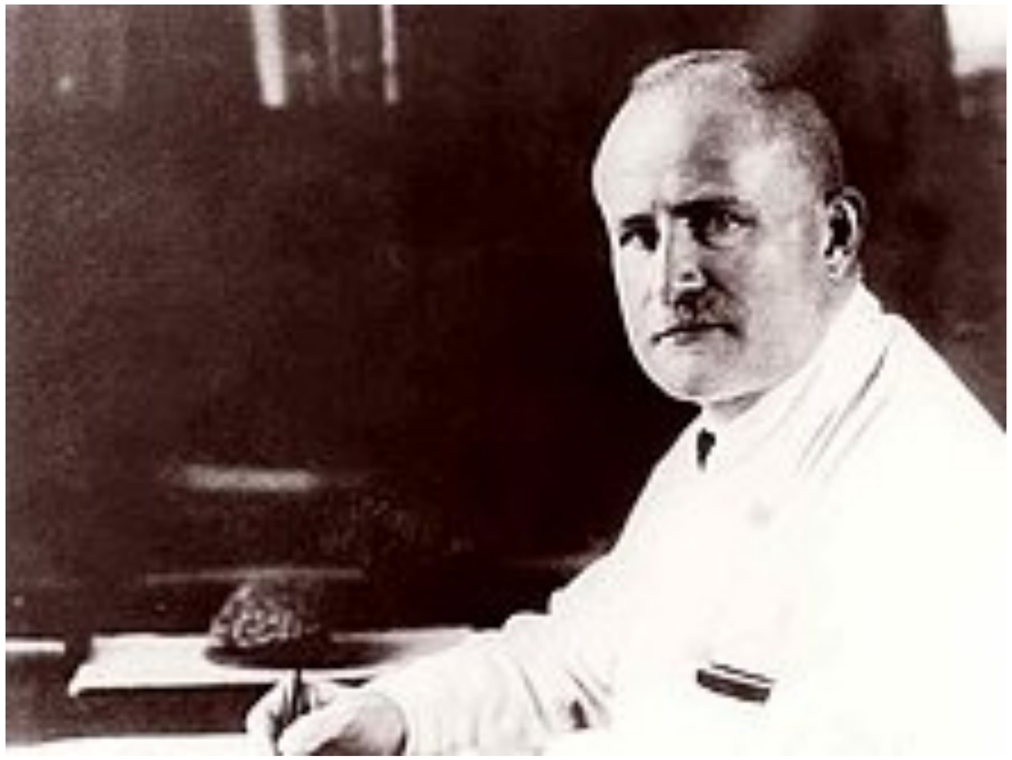
1893:

Mounted on a rearing and overturning horse in battery traveling in the valley of a narrow pass, I fell… and came to lie under the wheel of an artillery gun. At the last moment, the gun drawn by six harnessed horses stopped and I escaped with no more than a fright. This happened in the morning hours of a beautiful spring day. In the evening of the same day I received a telegraphed inquiry from my father asking how I was doing. It was the first and only time in my life that I received such an inquiry. My older sister, with whom I am in particularly intimate familial contact, had arranged this telegraphic inquiry since she suddenly told my parents that she distinctly knew that an accident had happened to me. My relatives lived in Coburg at that time. That was a case of spontaneous telepathy in which at the moment of mortal danger, envisioning certain death, I acted as sender and my sister, who was particularly close to me, acted as receiver.[8]

[Shure, Brain Waves, 2019]

Motivated (?) by his telepathic experience

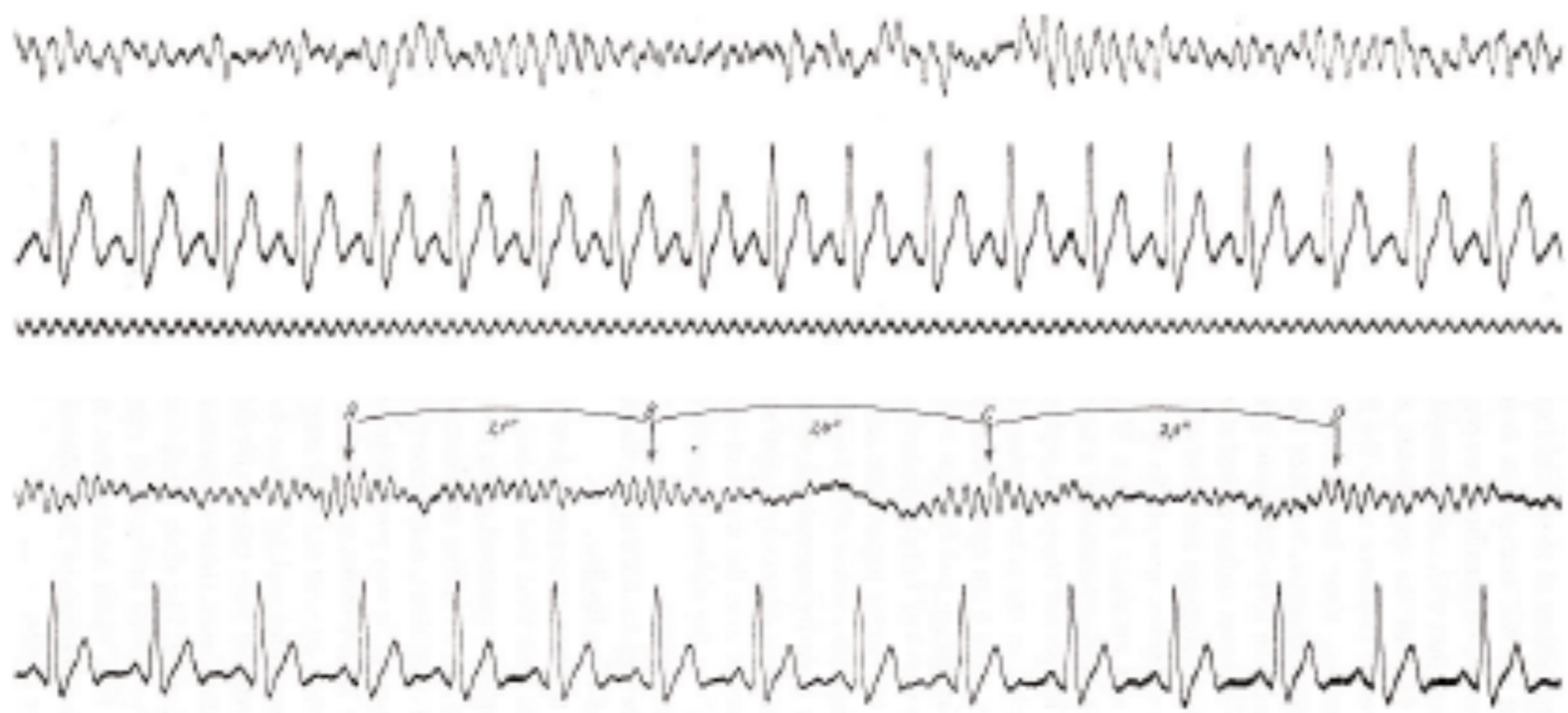
Brain rhythms



Hans Berger (1873-1941)

Inventor of the EEG
- first human brain voltage recording

1928-1929:



[Shure, Brain Waves, 2019]

Brain rhythms

Early days

[The Devil Commands, 1941]

https://www.youtube.com/watch?v=HMxmsPM1A8o see 2:00

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nature reviews neuroscience

Viev

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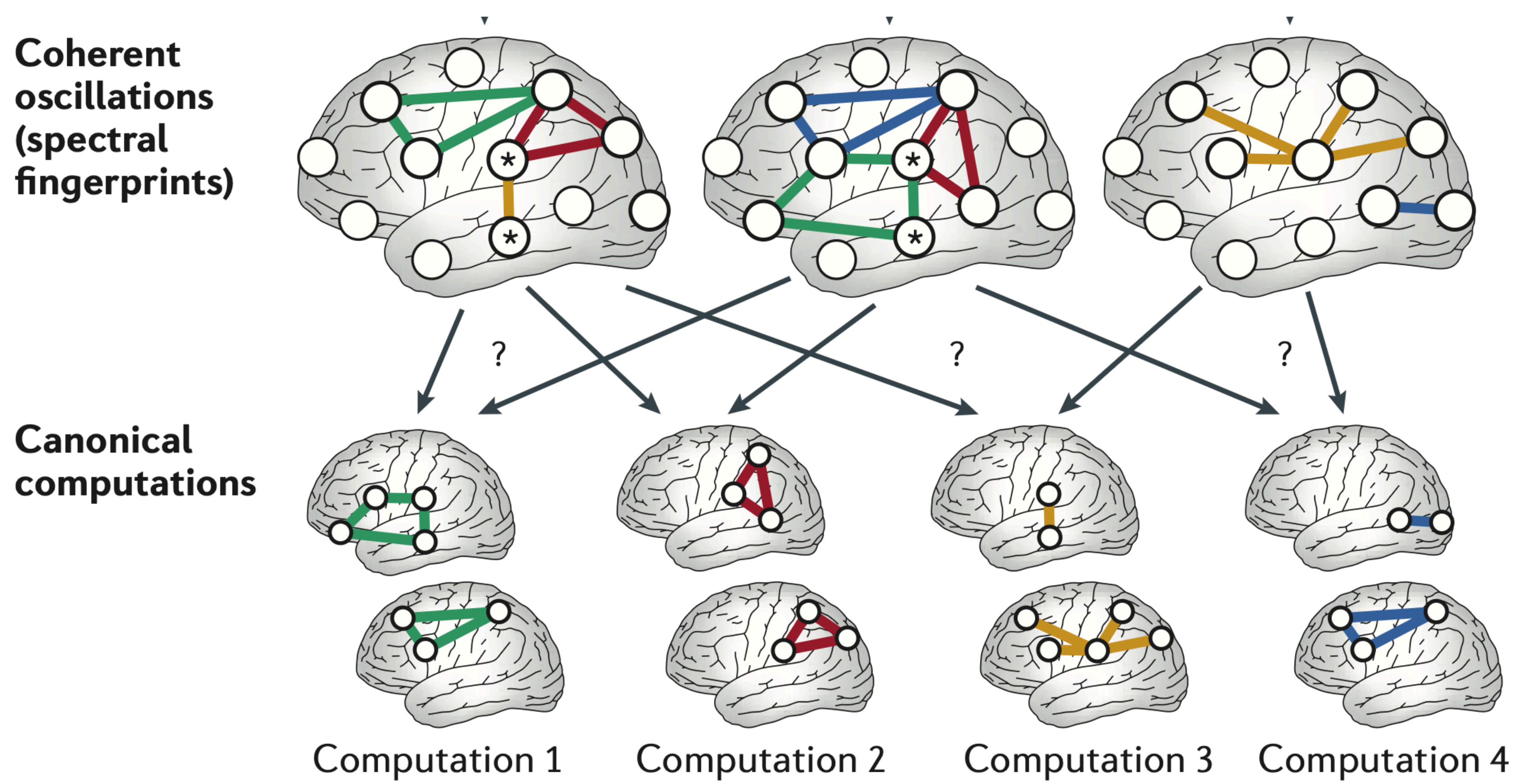
Review Article | Published: 11 January 2012

Spectral fingerprints of large-scale neuronal interactions

Markus Siegel, Tobias H. Donner & Andreas K. Engel

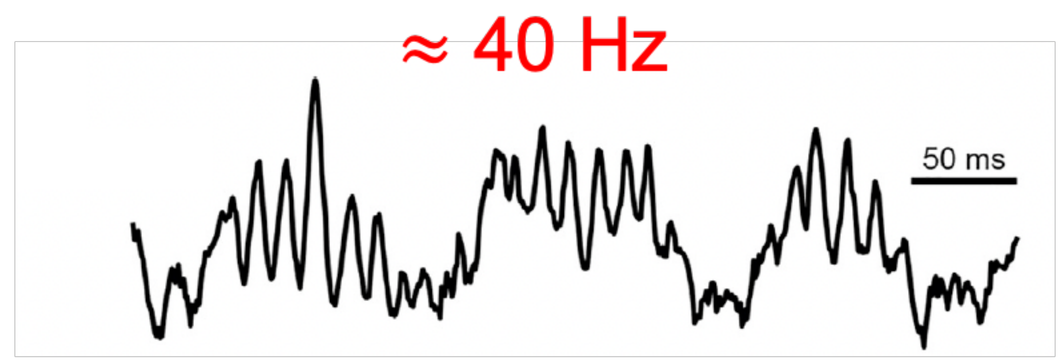
Nature Reviews Neuroscience **13**, 121–134 (2012) | Cite this article



Brain rhythms

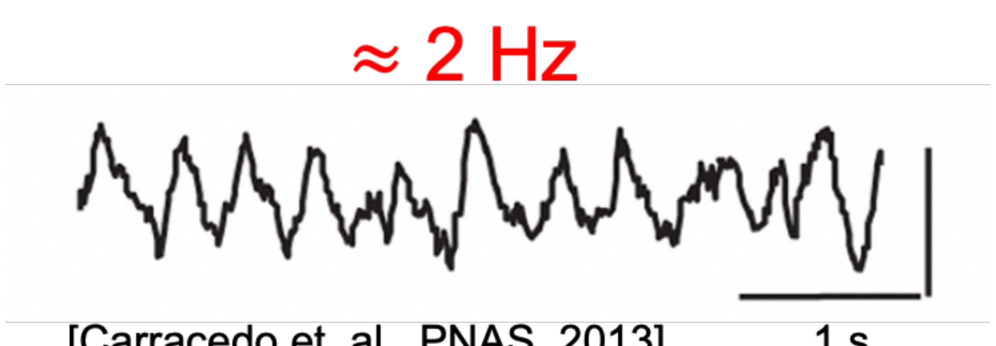
Fact: The brain can generate rhythmic activity.

[Buzsaki, Hippocampus, 2015]

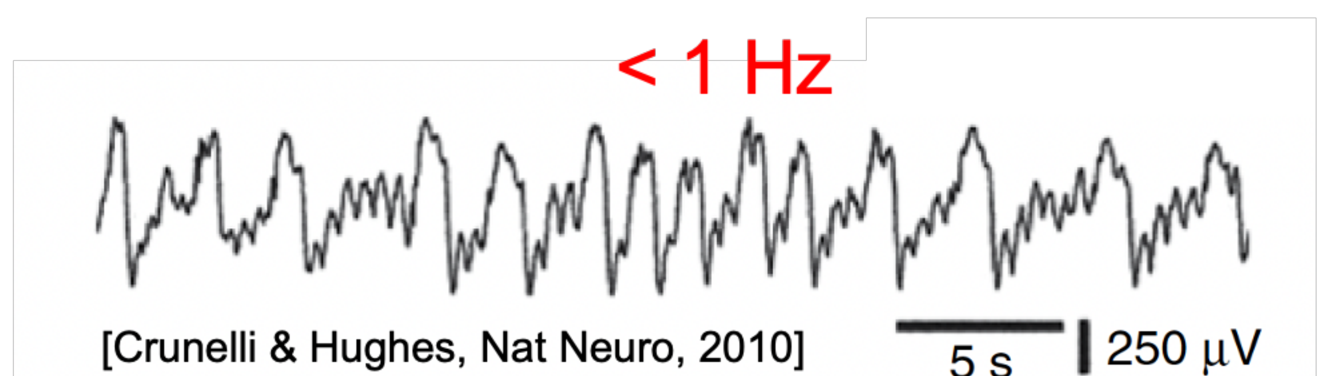


[Fernandez-Ruiz et al., Neuron, 2023]

[Kramer et. al., J Neurosci, 2021]



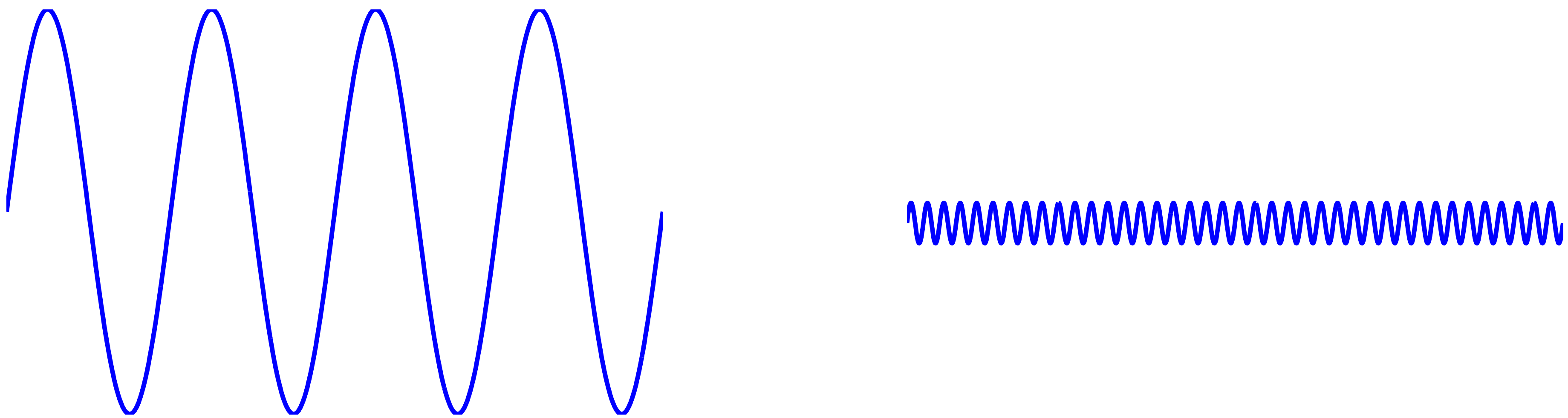
[Carracedo et. al., PNAS, 2013]



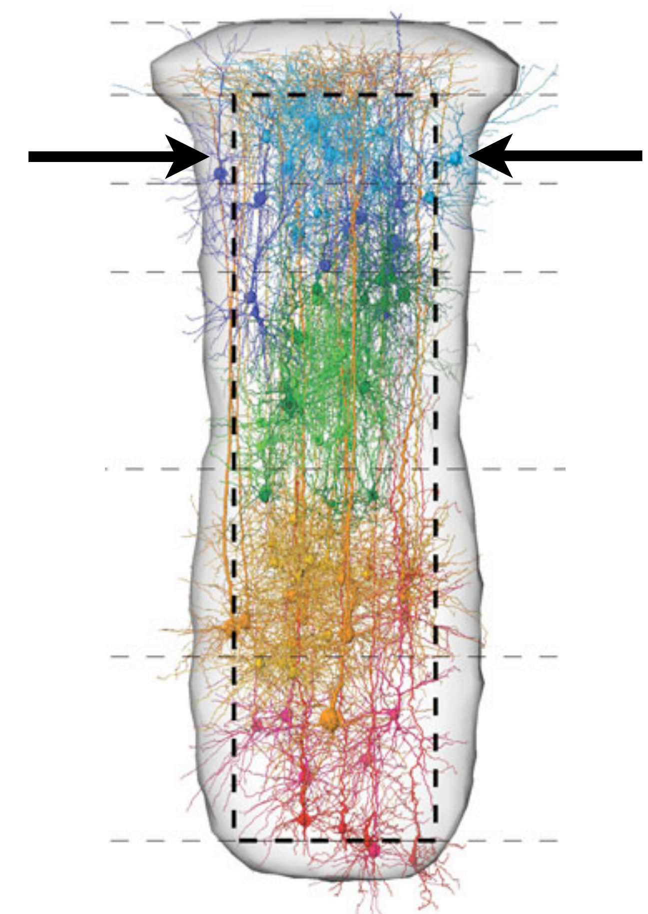
[Crunelli & Hughes, Nat Neuro, 2010]

Brain rhythms: "facts"

• Slower rhythms tend to be larger amplitude.



• The EEG ≈ generated by coordinated synaptic transmembrane currents of many neurons.



Very complicated.

Not completely understood.

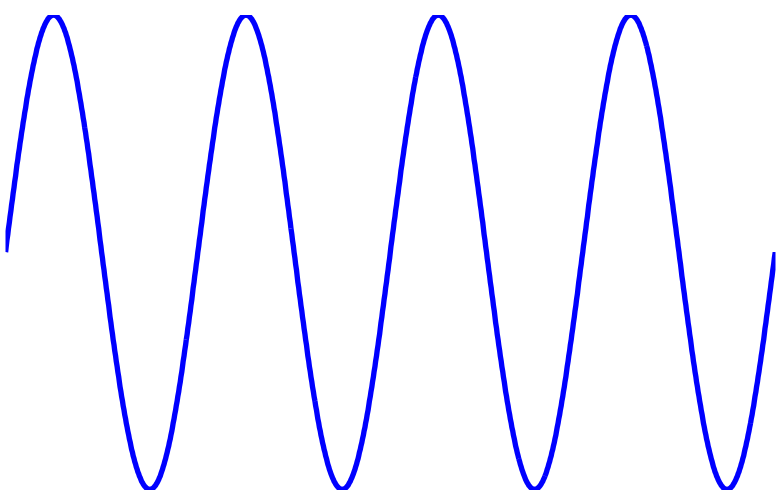
[Buzsáki et al. Nat Rev Neurosci (2012)]

Brain rhythms: "facts"

Rhythms indicate cortical arousal

–Modulate the firing patterns of neurons.

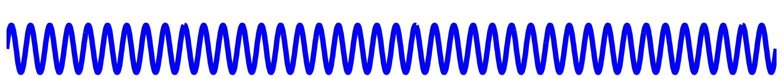
Low frequency, high amplitude: low cortical arousal



Neural activities are . . . synchronized.

Groups of cells act in concert = large EEG signal.

High frequency, low amplitude: high cortical arousal



Neural activities are . . . desynchronized.

Groups of cells involved in separate activities = small EEG signal.

Note: A healthy brain is a desynchronized brain.

Brain rhythms: characterization

Q: How do we characterize these rhythms?

To start, we can visualize and describe the rhythms.

Typical features:

Amplitude: Large or small ?

Frequency: Fast or slow ?

Shape: Sinusoidal, square, triangle, . . . ?

Duration: Long or short lasting ?

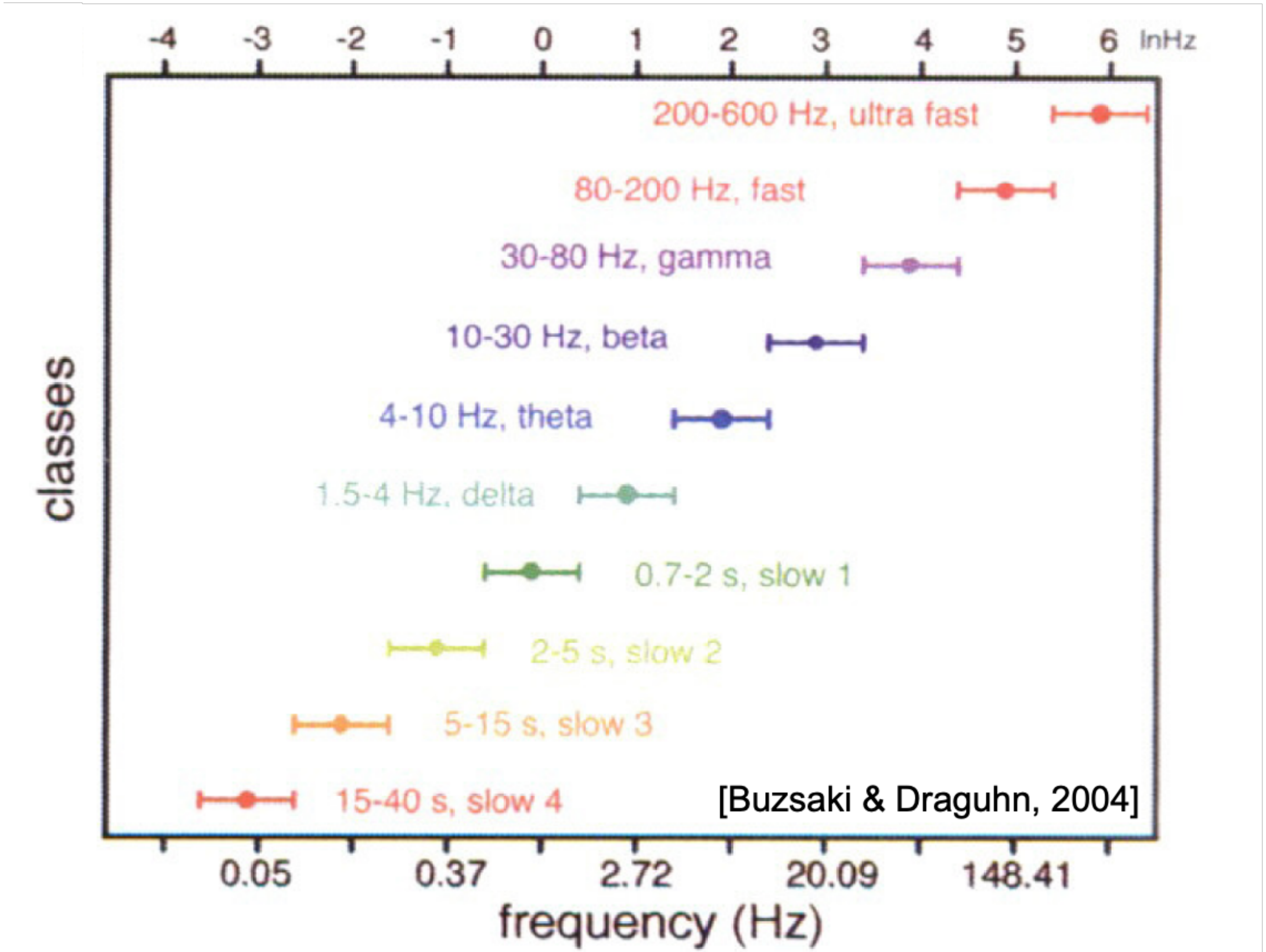
Our focus (usually) is frequency: How fast or slow is a rhythm?

Q: Why? Because different frequency rhythms are associated with different functions ...

Brain rhythms and functions

Many frequency bands, each associated with different functions.

Ex:

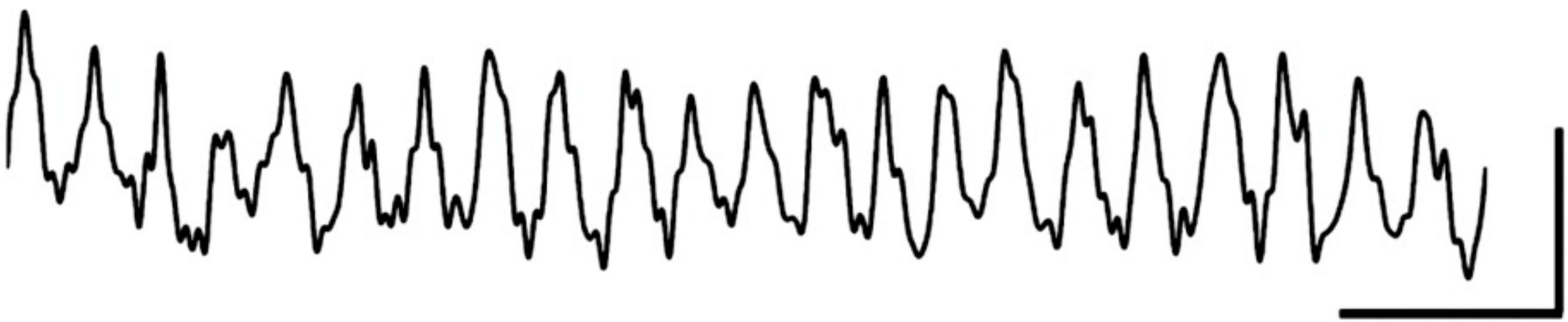


Let's look more carefully at some of these frequency bands . . .

Brain rhythms: theta

Theta: 4-8 Hz

Note: Theta frequency range different; the borders of ranges are not exact.

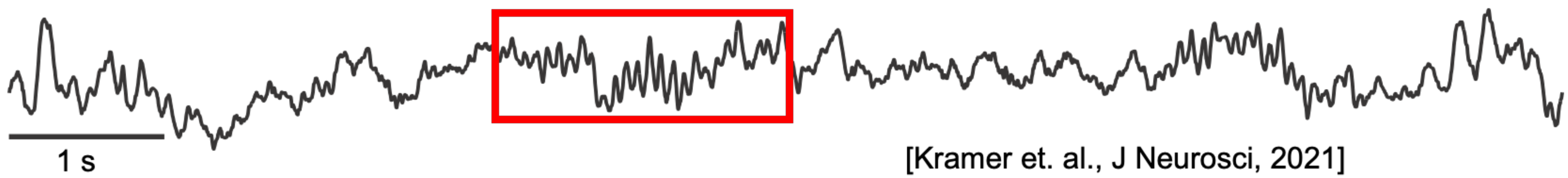


Function: Not completely understood.

In rats: learning and memory
location
motor behavior
sleep
emotional arousal
fear conditioning

Brain rhythms: alpha

Alpha: 8-12 Hz Note: This band not in Slide #7 !



[Kramer et. al., J Neurosci, 2021]

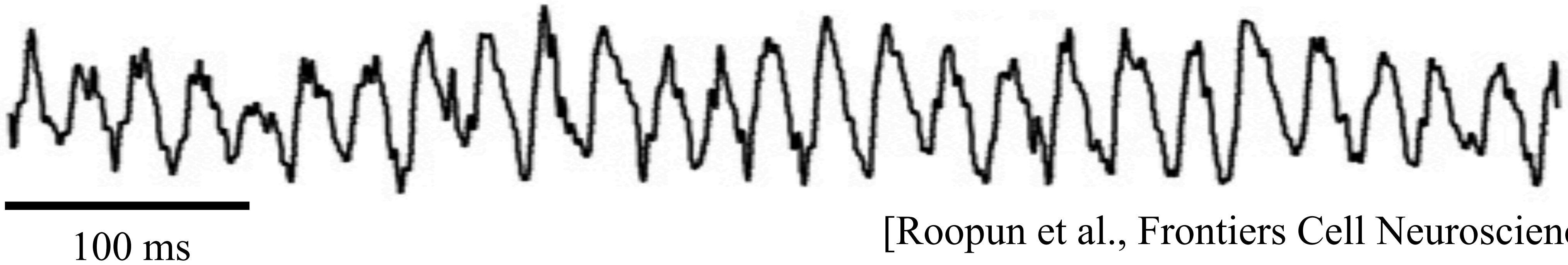
- The first EEG wave studies [Berger 1931]
- In EEG, strongest above occipital lobes when eyes closed at rest.

Function: "idling rhythm" - alert but still brain state
cortical operations in the absence of sensory inputs
disengagement of task-irrelevant brain areas

However, alpha also associated with attention, sensory awareness, memory consolidation.

Brain rhythms: beta

Beta: 12-30 Hz



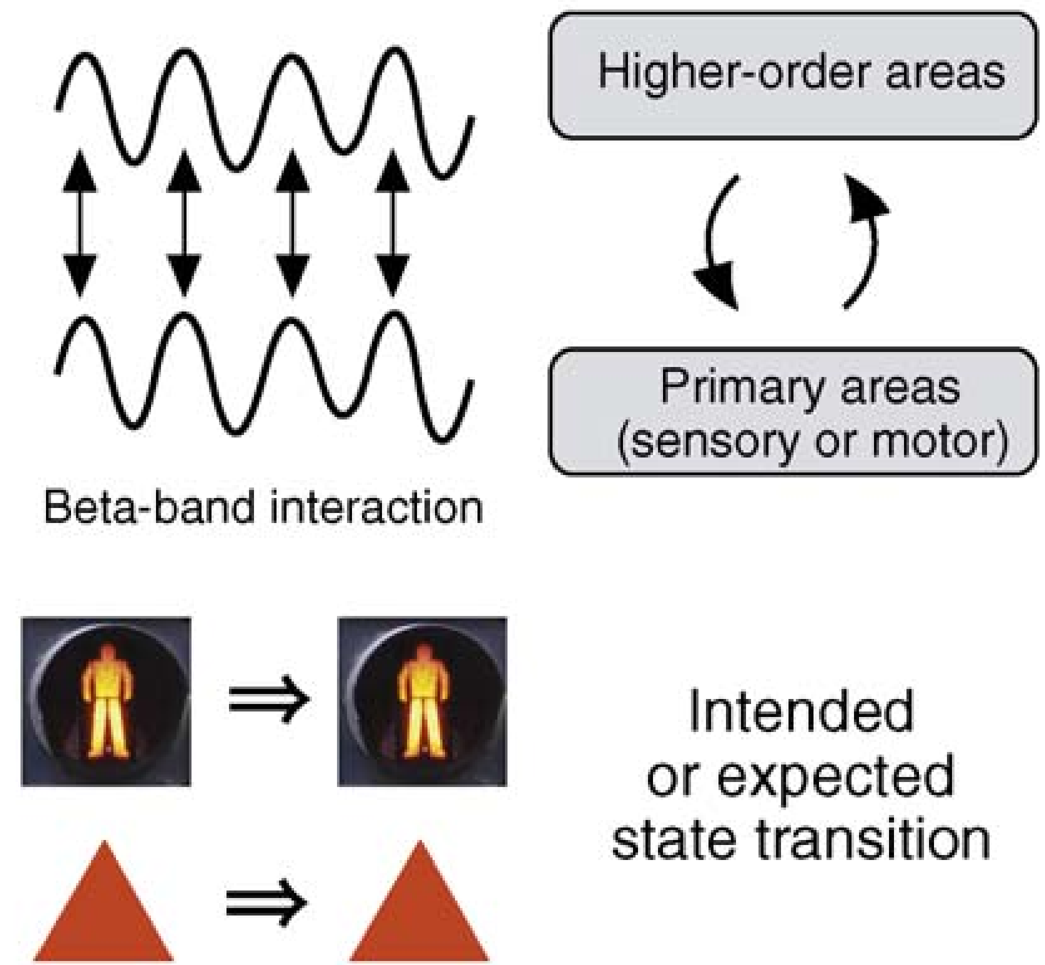
[Roopun et al., Frontiers Cell Neuroscience, 2008]

Function: Not fully understood.

motor functions (e.g., steady-state contractions).

“Maintenance of the status quo”

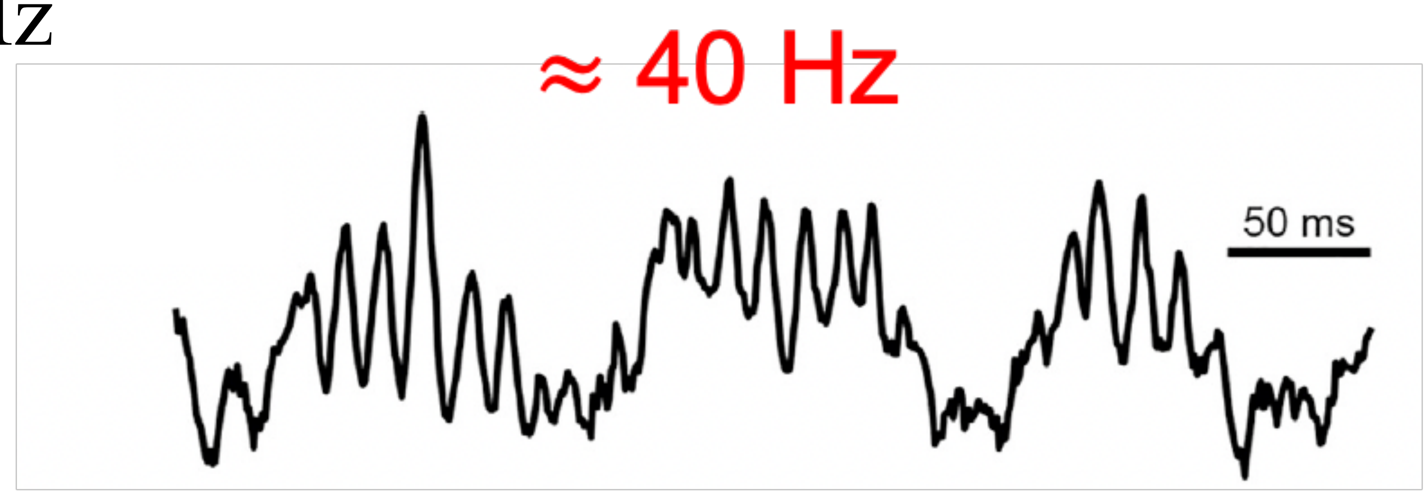
“Couple” distant brain regions.



[Engel, Fries 2010]

Brain rhythms: gamma

Gamma: 30-50 Hz



[Fernandez-Ruiz et al., Neuron, 2023]

Function: Associated with a broad range of processes:
"binding"
attention
movement preparation
memory formation
conscious awareness

Note: Typically hard to see in EEG.   **Q:** Why?

Brain rhythms: Other bands

There are many other frequency bands . . .

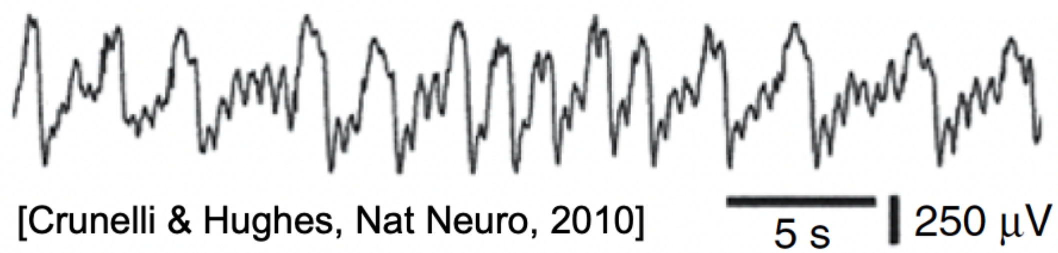
Slower

–**Delta**: 1-4 Hz

Sleep, learning, motivation

–**Slow cortical potential**: < 1 Hz

Emergence of consciousness?



[Crunelli & Hughes, Nat Neuro, 2010]

Faster

– **High gamma**: 50-120 Hz

Coordination of neural activity

– **Ripples, HFO, UFO**: > 120 Hz

Replay of memories,
onset of seizures . . .

[Buzsaki, Hippocampus, 2015]

Brain rhythms in disease

Rhythms are sometimes associated with pathologies.

Ex: Seizure

[Martinet et al., Nat Comm, 2017]

Q: What rhythms do we see?

HFO Beta Alpha Theta Delta

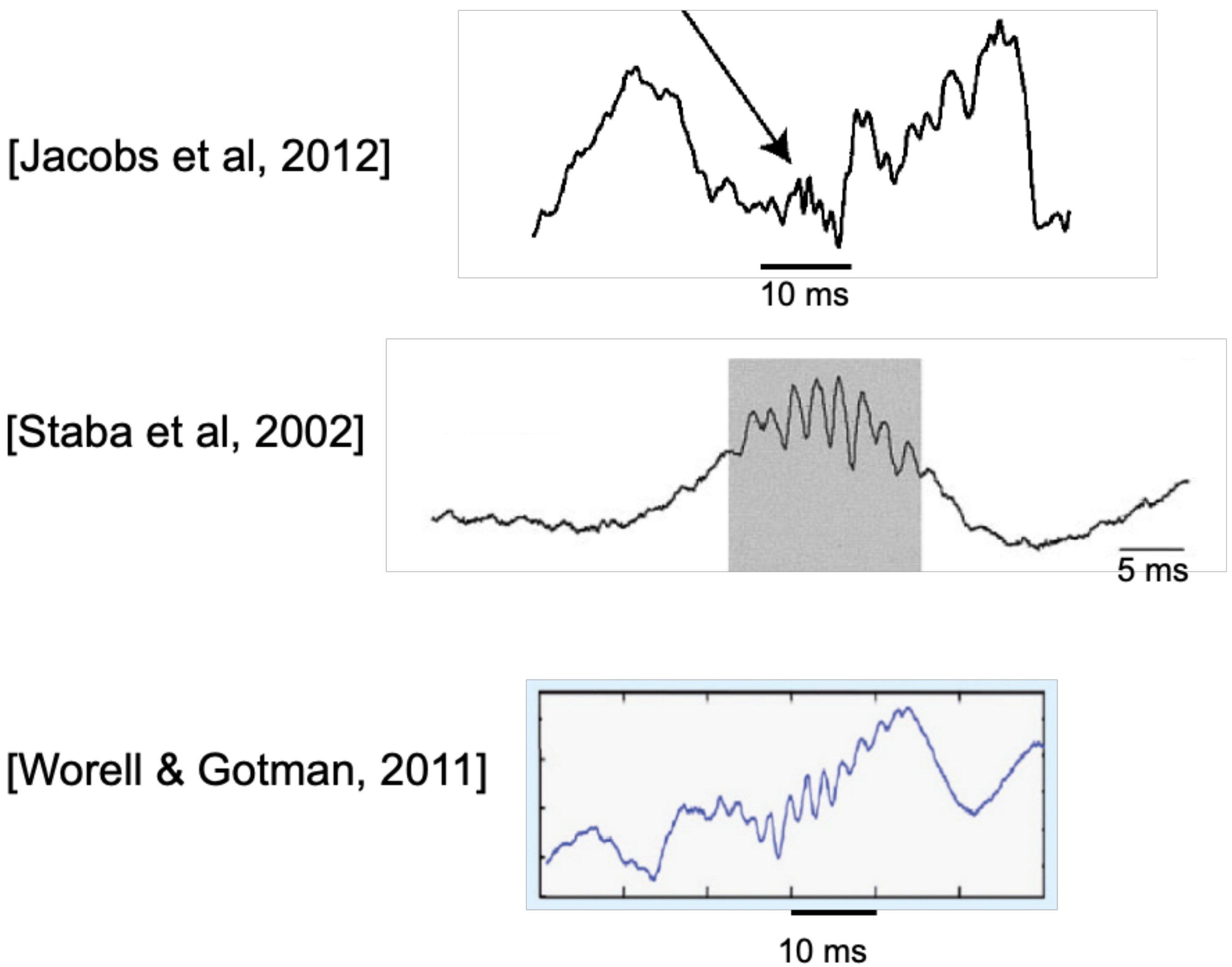
Q: How does the amplitude change?

Note: Used the band labels, but "function" very different.

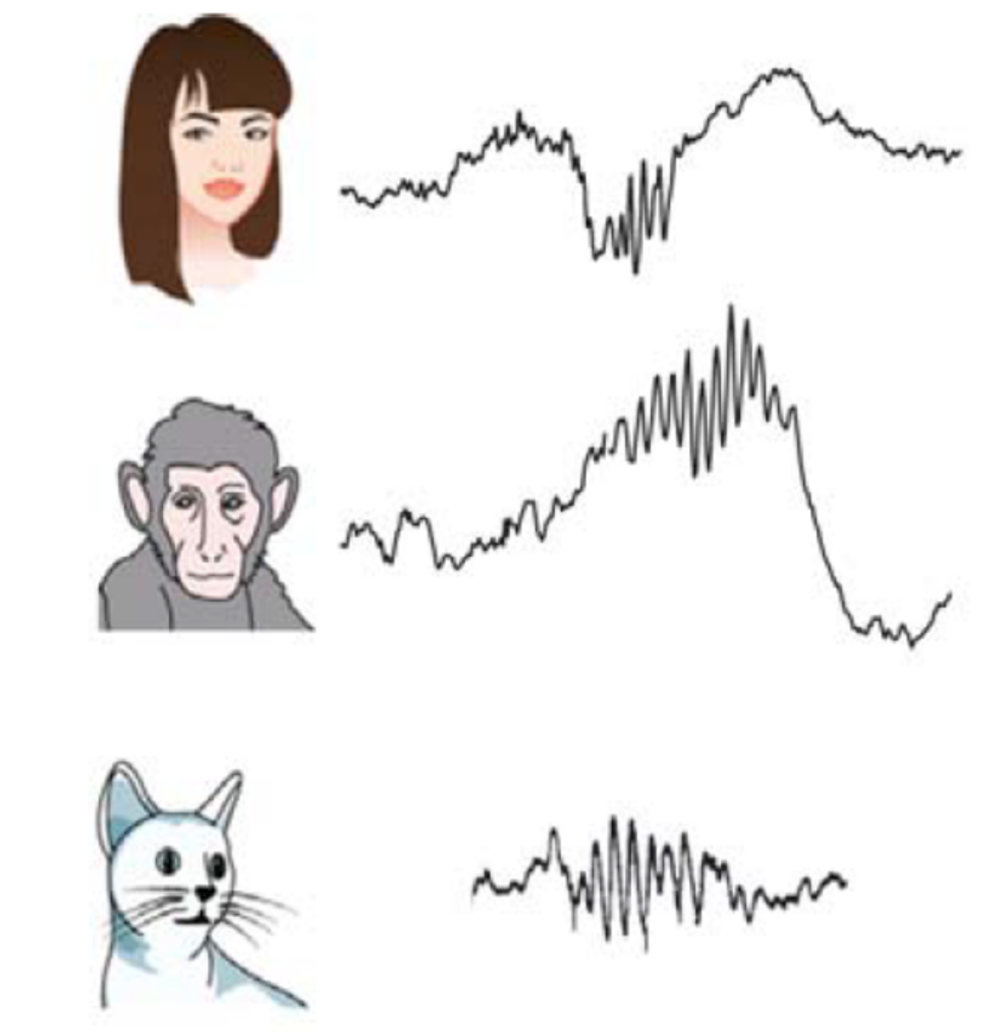
Brain rhythms in disease

Rhythms are sometimes associated with pathologies.

Pathological ripples



Physiological ripples



[Buzsáki, Hippocampus, 2015]

<u>Note</u>: Used the band label, but “function” very different.

Brain rhythms are meaningless . . .

H: Brain rhythms are epiphenomena.

Large buildings: sway in the wind.

These oscillations are not performing a function, in fact they're unwanted.

Q: Is the same true in the brain? Oscillations are an echo of some underlying function?

Q: Why so many oscillations?

Big questions

Q: Why do we observe rhythms in the brain?

–Functional role

–Epiphenomena

Q: What mechanisms support rhythms?

–Biological

–Dynamical

Q: Why do rhythms interact?

–Mechanisms

–Functions

–Measures

Answers?
Data analysis and models ...

Q: Could you build *intelligence* without rhythms?

Characterize brain rhythms

Many sophisticated tools to do so.

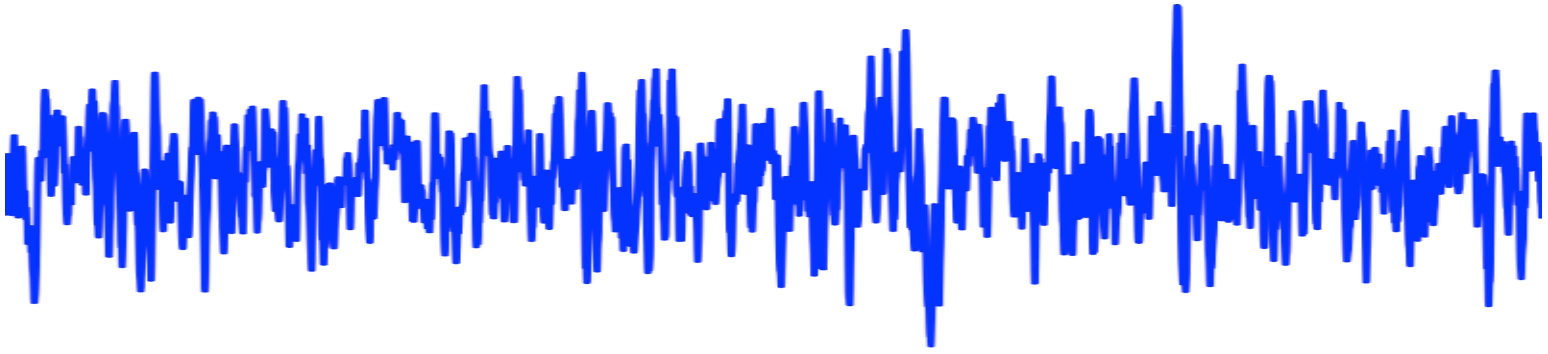
Today, consider two:

–Visual inspection

–Intuition for the spectrum

Idea:

Visual inspection: Plot the data. What do you see?



Spectrum: Break down the data into sinusoids . . .

Remember: sinusoids . . .

$$V[t] = A\cos(2\pi f t)$$

Voltage as a function of time

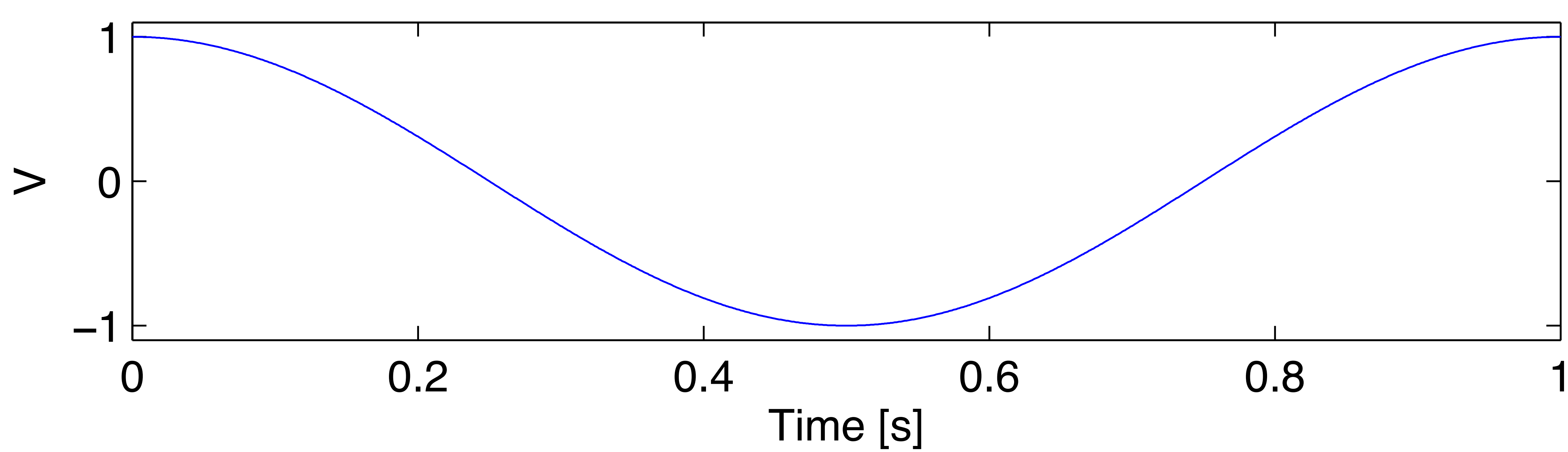
Amplitude

Frequency [Hz]

Time [s]

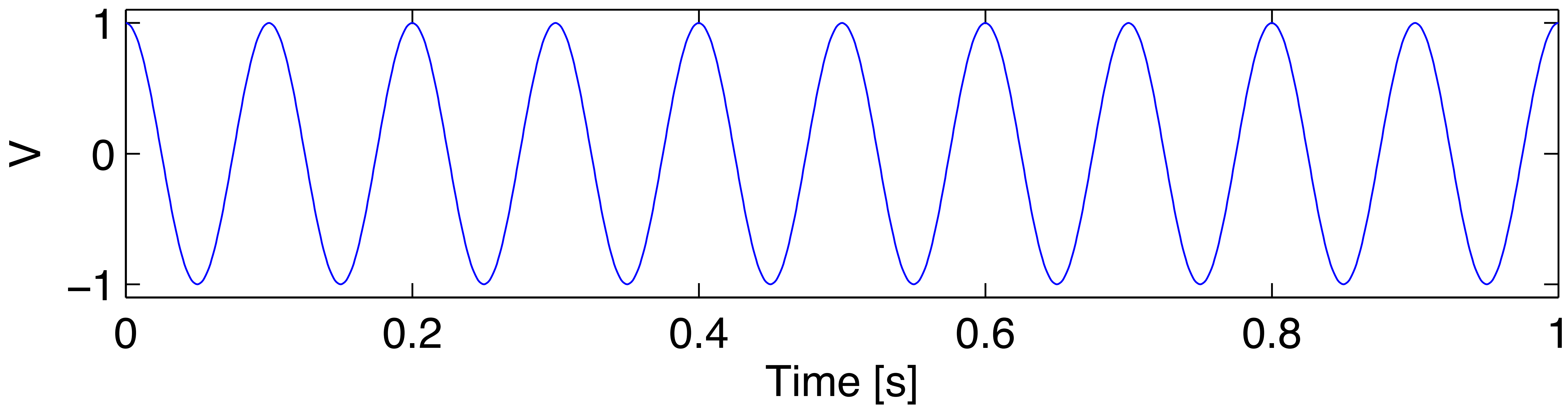
Ex. Consider: f = 1 Hz

Q: What does it look like?



Remember: sinusoids . . .

Q: What is the frequency of this sinusoid?



A: 10 cycles in 1 second, so 10 Hz.

Note: Visual inspection is often useful.

Remember: sinusoids . . .

In addition to cosine, there's also sine:

$$V[t] = B\sin(2\pi f t)$$

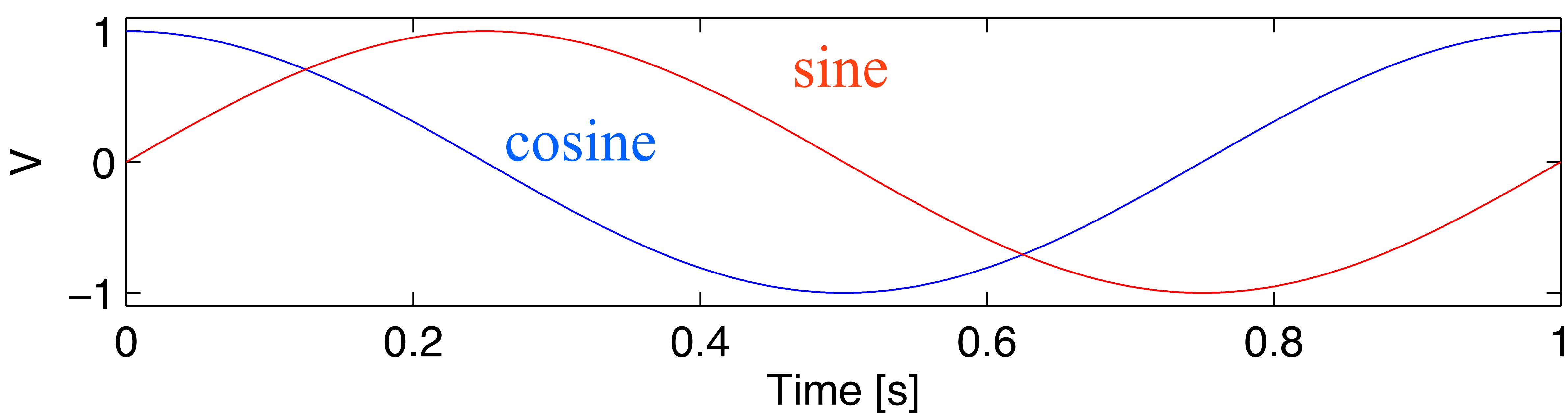
Voltage as a function of time

Amplitude

Frequency [Hz]

Time [s]

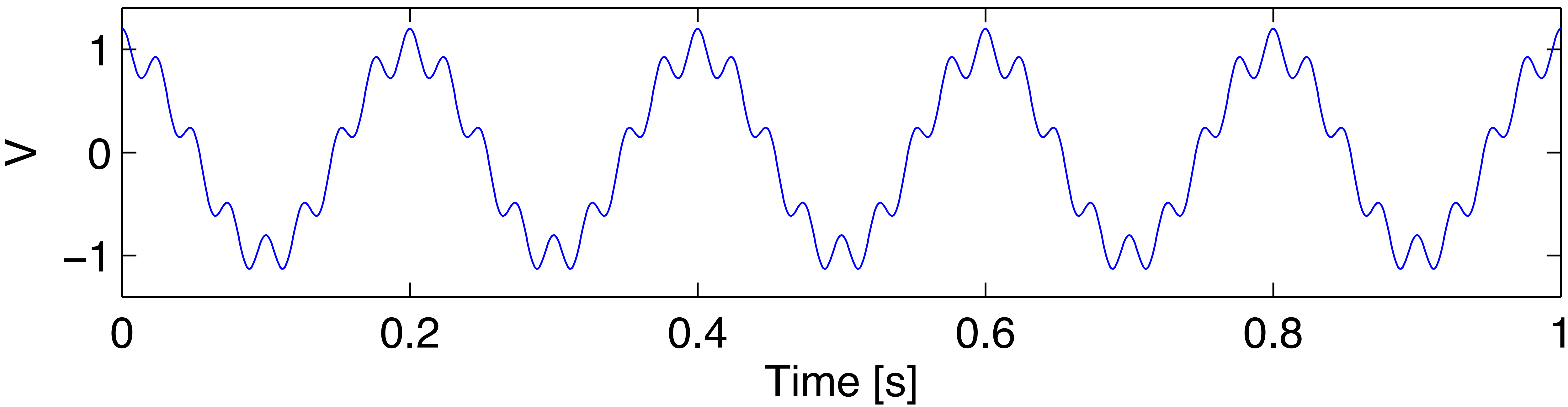
Ex. Consider: f = 1 Hz



Q: What's the difference?

Example: rhythmic signal

Consider the signal below:



Q: What are the rhythms?

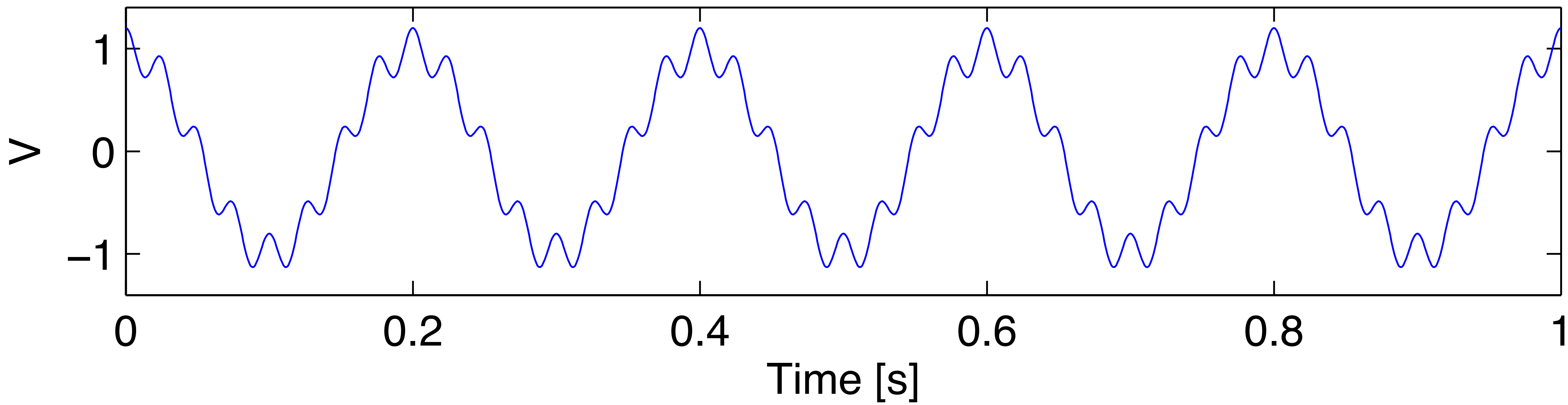
A: Apply visual inspection . . . Slow and fast

5 Hz 40 Hz

Q: What has larger amplitude?

Example: rhythmic signal

So, we can represent this signal . . .



. . . as the sum of two sinusoids:

A slow, large amplitude sinusoid + a fast, small amplitude sinusoid

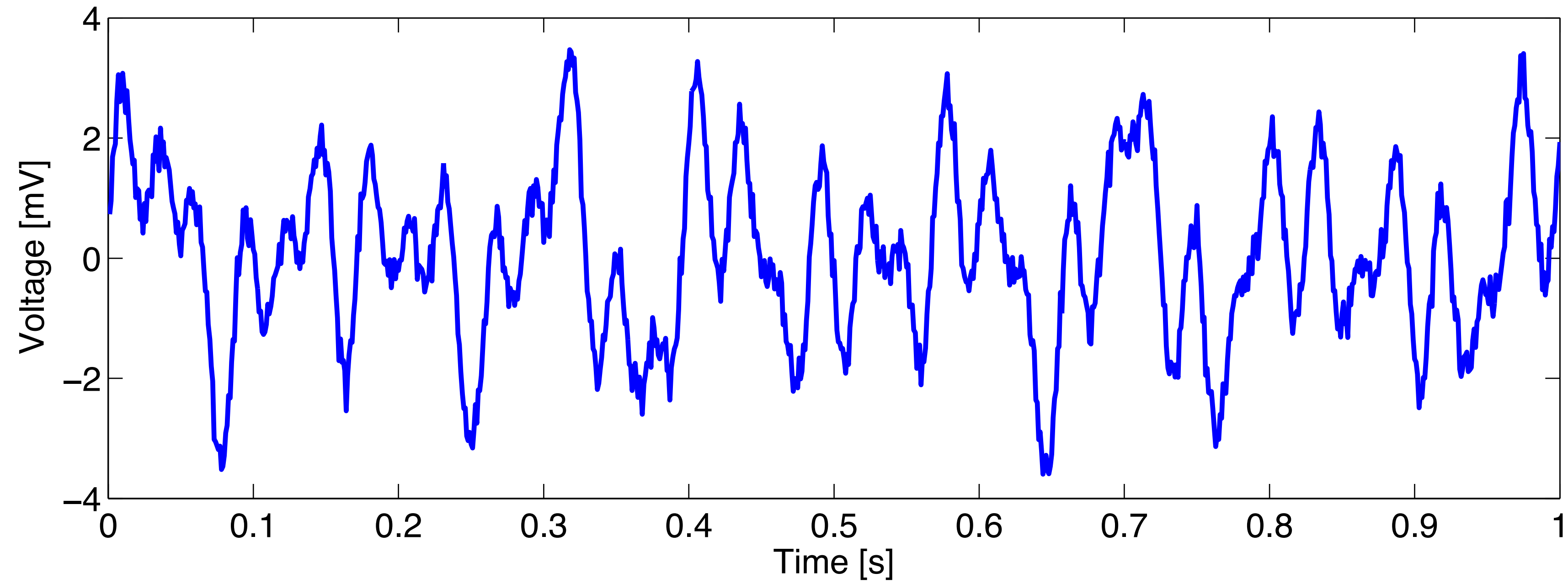
$$V[t] = \underset{1.0}{A_1} \cos(2\pi \underset{5\ \text{Hz}}{f_1} t) + \underset{0.1}{A_2} \cos(2\pi \underset{40\ \text{Hz}}{f_2} t)$$

We get a <u>simpler</u> representation of the signal. That's the idea of the spectrum.

Idea: Spectrum

Consider:

V =



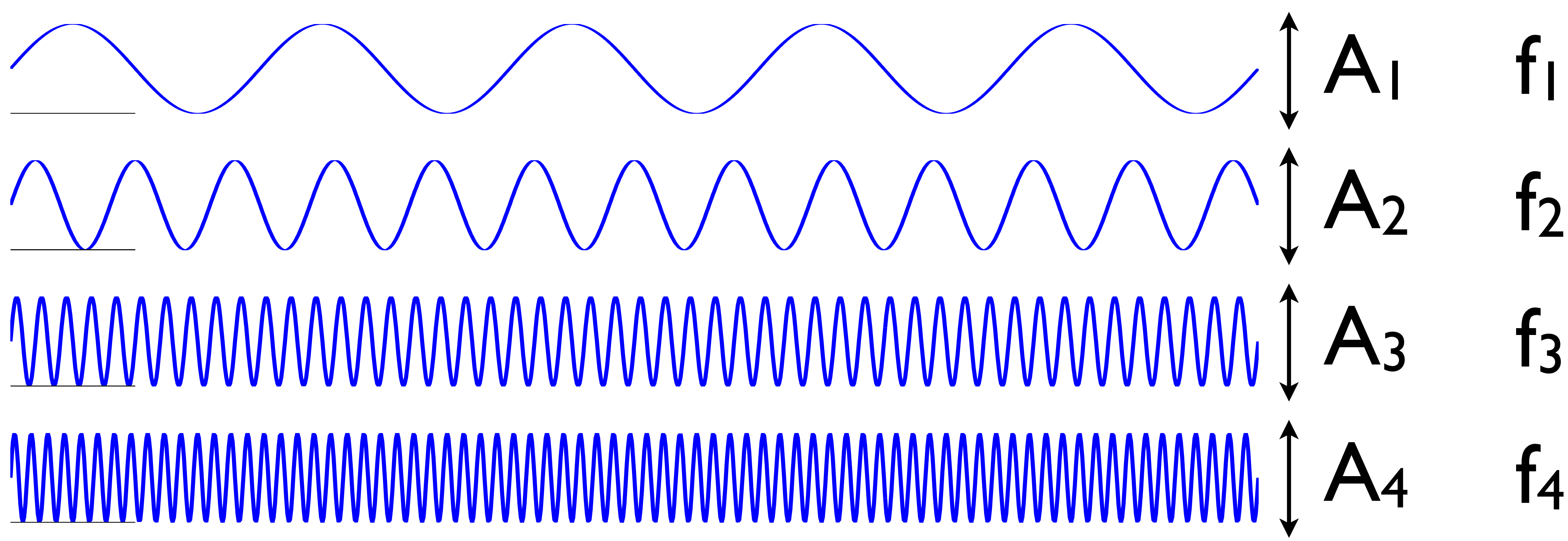
- Decompose signal into oscillations at different frequencies.

V =

+

+

+



+ . . .

Represent V as a sum of sinusoids (e.g., part 7 Hz, part 10 Hz, . . .)

Idea: Spectrum

So, in equations:

$$V[t] = A_1 \cos(2\pi f_1 t) + B_1 \sin(2\pi f_1 t) + A_2 \cos(2\pi f_2 t) + B_2 \sin(2\pi f_2 t) + \ldots$$

amplitude frequency amplitude frequency

Or, more generally:

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

sum over many frequencies

amplitude

oscillation at frequency f_j

<u>Note</u>: A_j and B_j can be zero.
(some rhythms make no contribution to V[t])

<u>Note:</u> A_j and B_j are large when f_j is a good match to the data.

Aside: Spectrum

Note: Think of sine & cosine as accounting for **phase**.

$$\underset{\text{amplitude}}{C_j} \cos(2\pi \underset{\text{freq}}{f_j} t + \underset{\text{phase}}{\phi_j})$$

Aside: $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$

$$= \boxed{C_j}\cos(2\pi f_j t)\boxed{\cos(\phi_j)} - \boxed{C_j}\sin(2\pi f_j t)\boxed{\sin(\phi_j)}$$

$$A_j \qquad\qquad B_j$$

$$= \boxed{A_j}\cos(2\pi f_j t) + \boxed{B_j}\sin(2\pi f_j t)$$

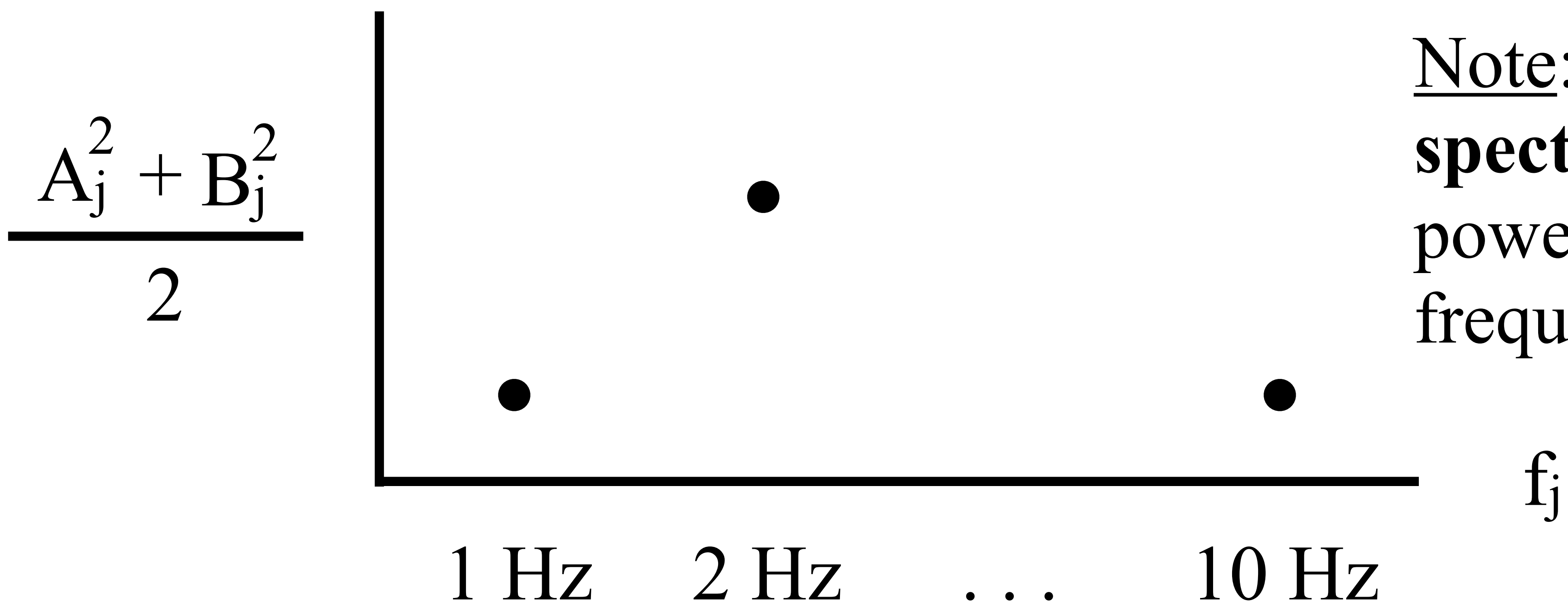
Decompose V[t] into sine/cosine or amplitude/phase.

Plot: Spectrum

We can represent these amplitudes and frequencies graphically:

Plot: $\dfrac{A_j^2 + B_j^2}{2}$ versus f_j for each j

<u>Note</u>: The summed amplitudes squared.
Called the "power"

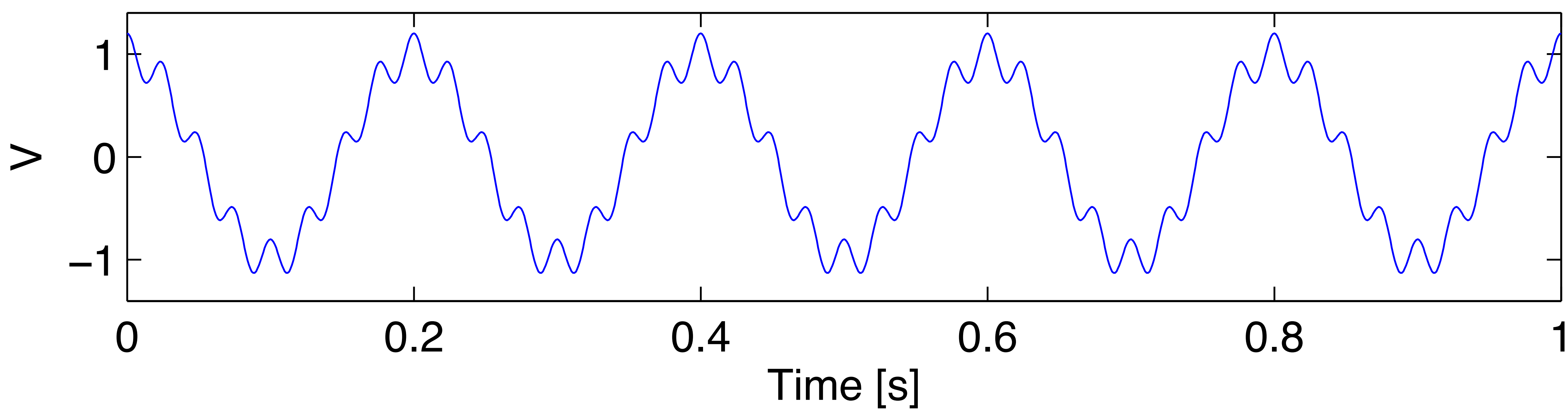


<u>Note</u>: This is called the **spectrum**; a plot of power versus frequency

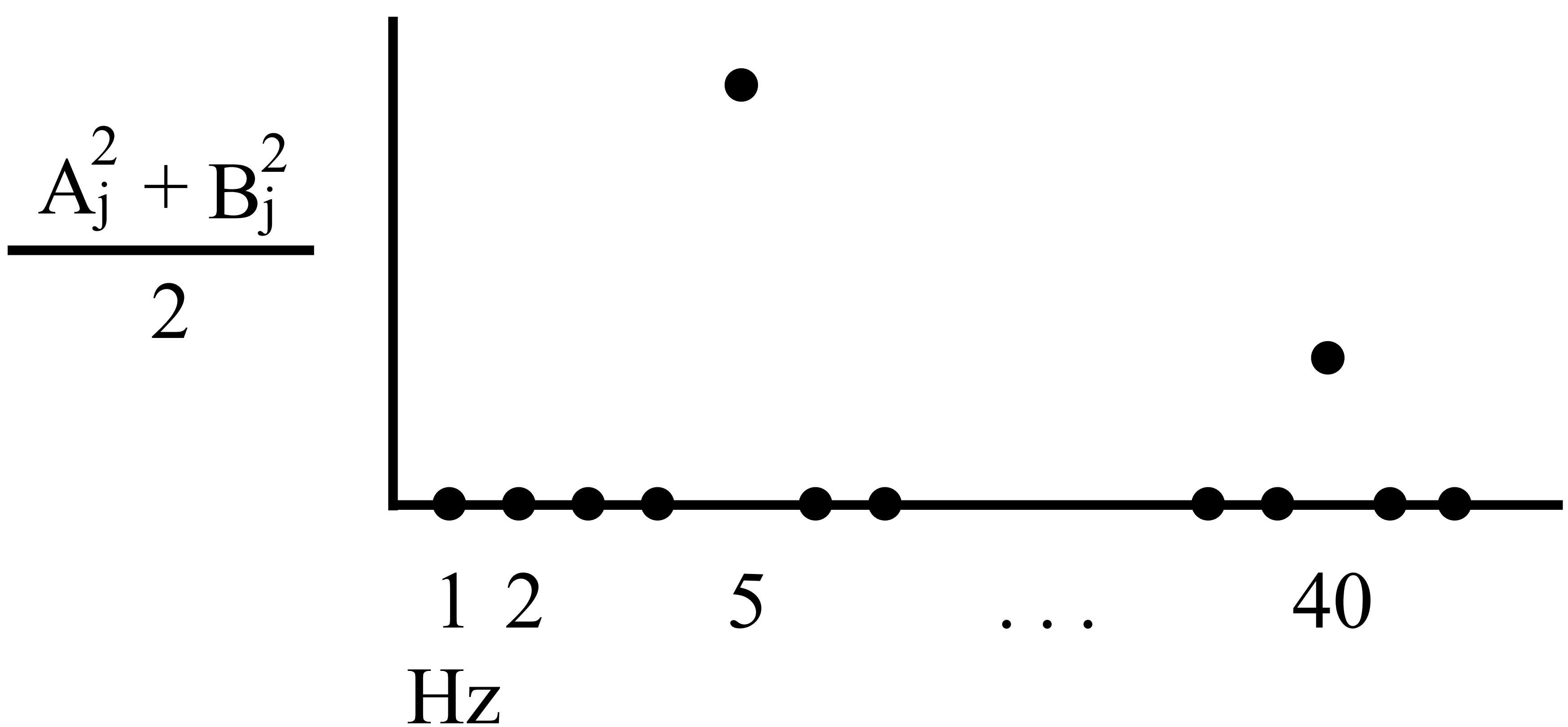
The peaks represent the <u>dominant rhythms</u> in the signal.

Example: rhythmic signal

Ex:



Plot the spectrum:

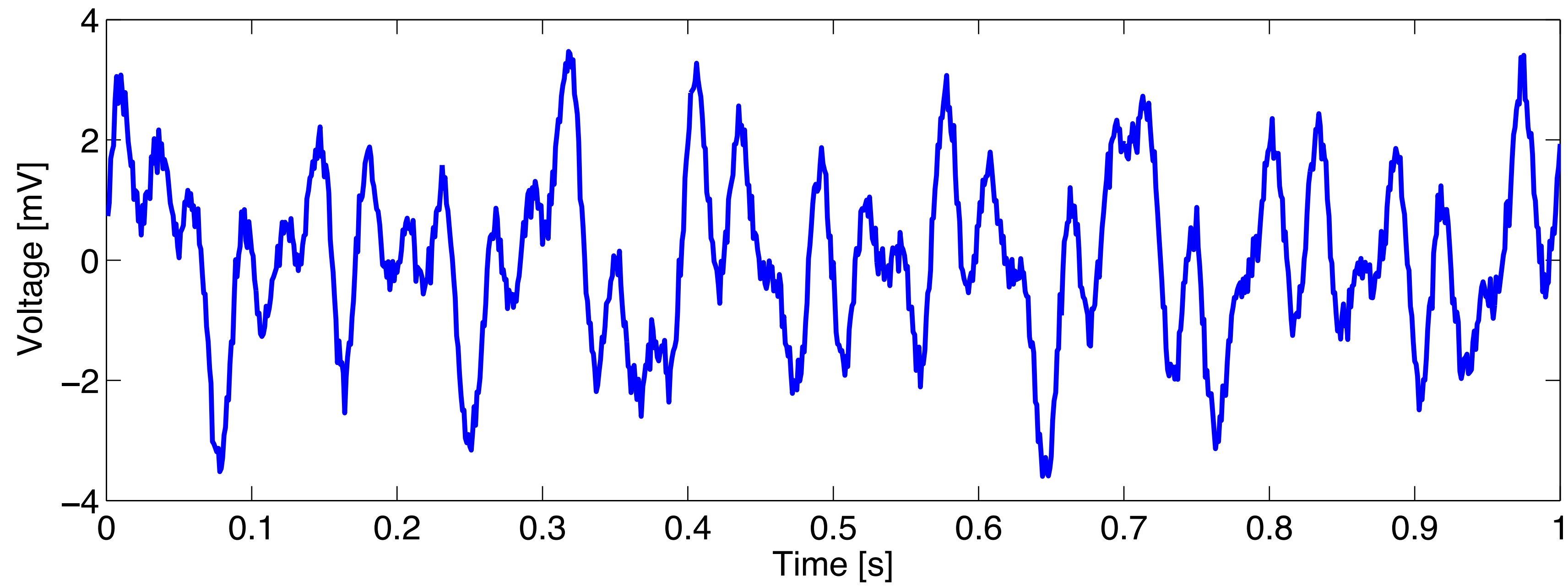


Note: The peaks correspond to the dominant rhythms

Example: rhythmic signal

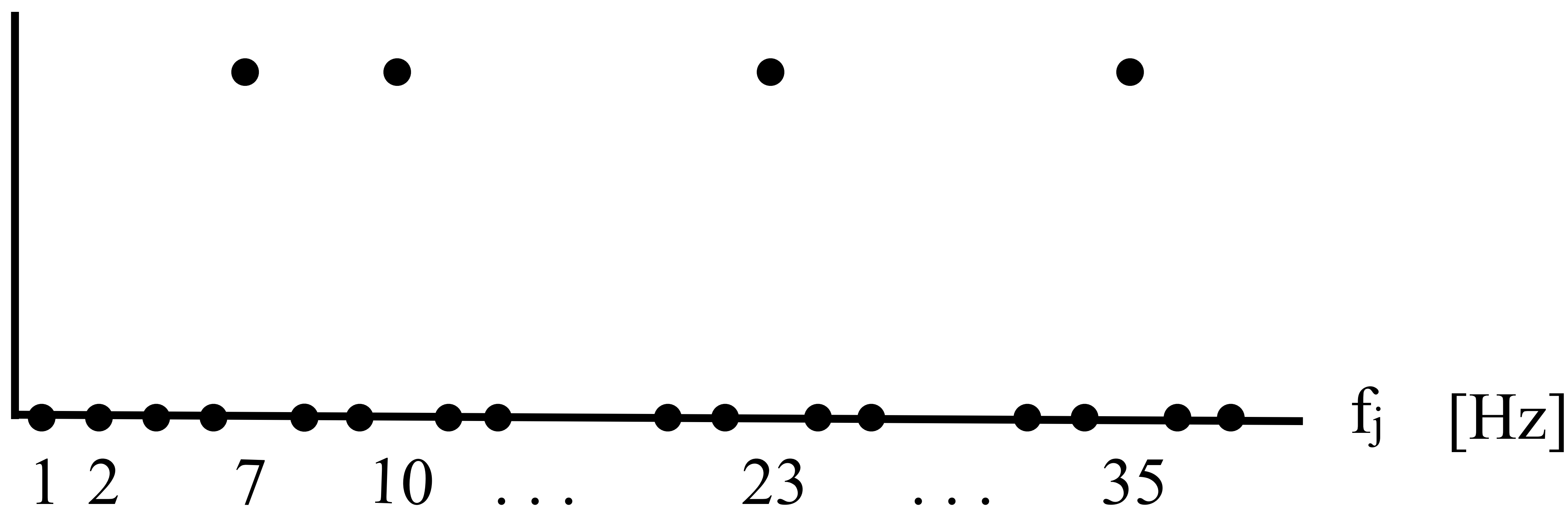
Ex.

V =



Note: It's complicated   Plot the spectrum:

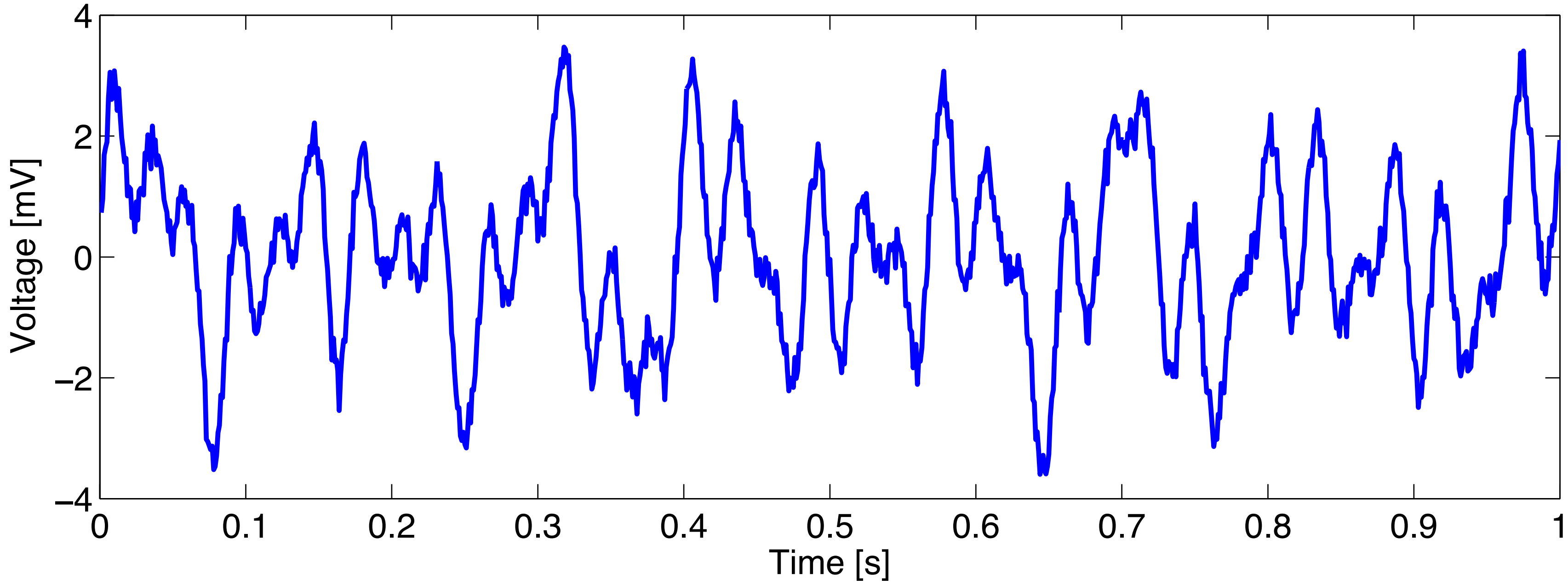
$$\frac{A_j^2 + B_j^2}{2}$$



Q: What's happening here?

Example: rhythmic signal

So, by computing the power spectrum, we find the complicated signal:

V = 

We find it's the sum of 4 sinusoids at frequencies:

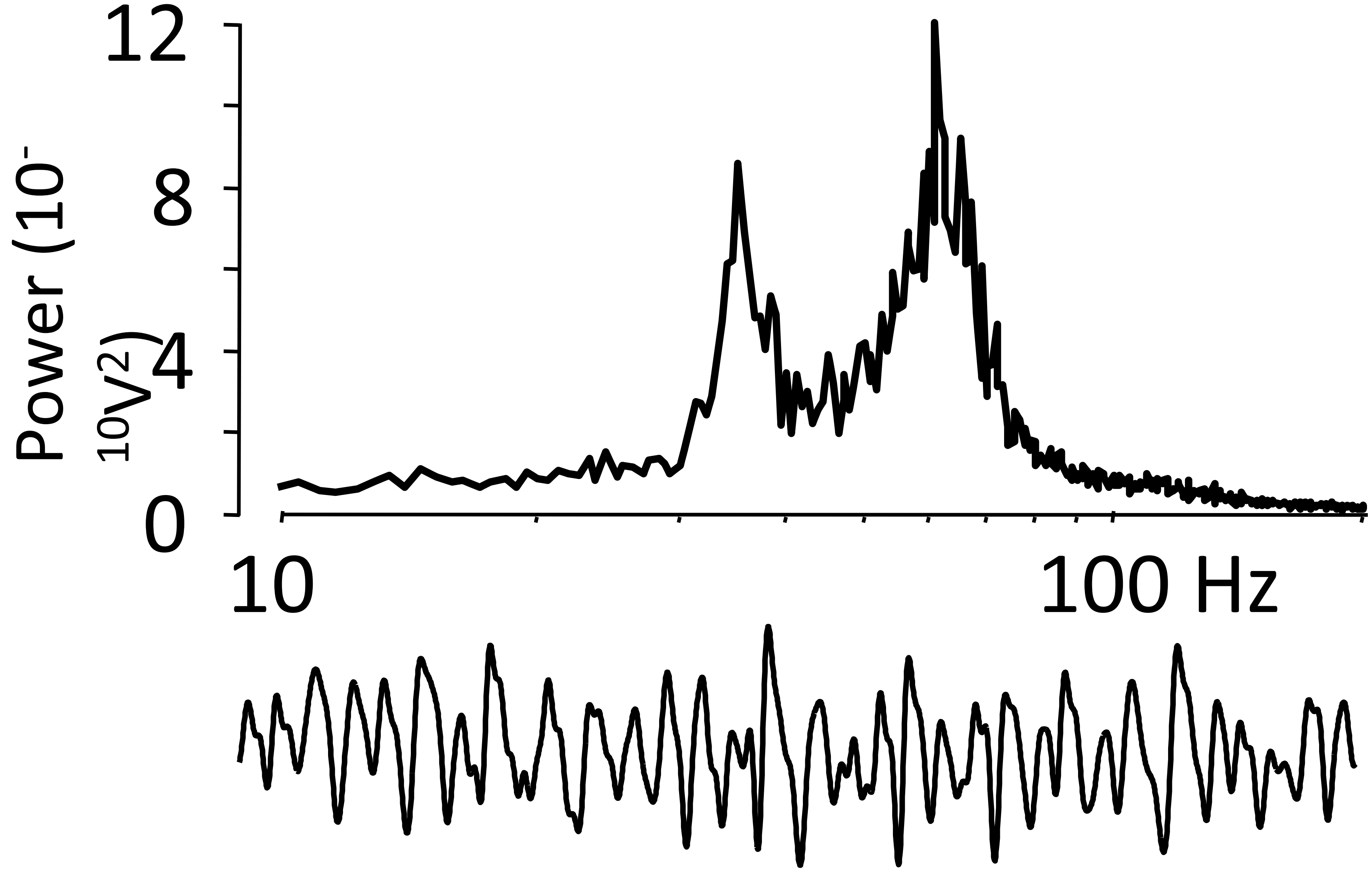
7 Hz, 10 Hz, 23 Hz, and 35 Hz

A much simpler representation of brain activity.

Example: Real world

Q: What does the power spectrum of real-world brain signals look like?

Ex. From a slice of rat cortex:



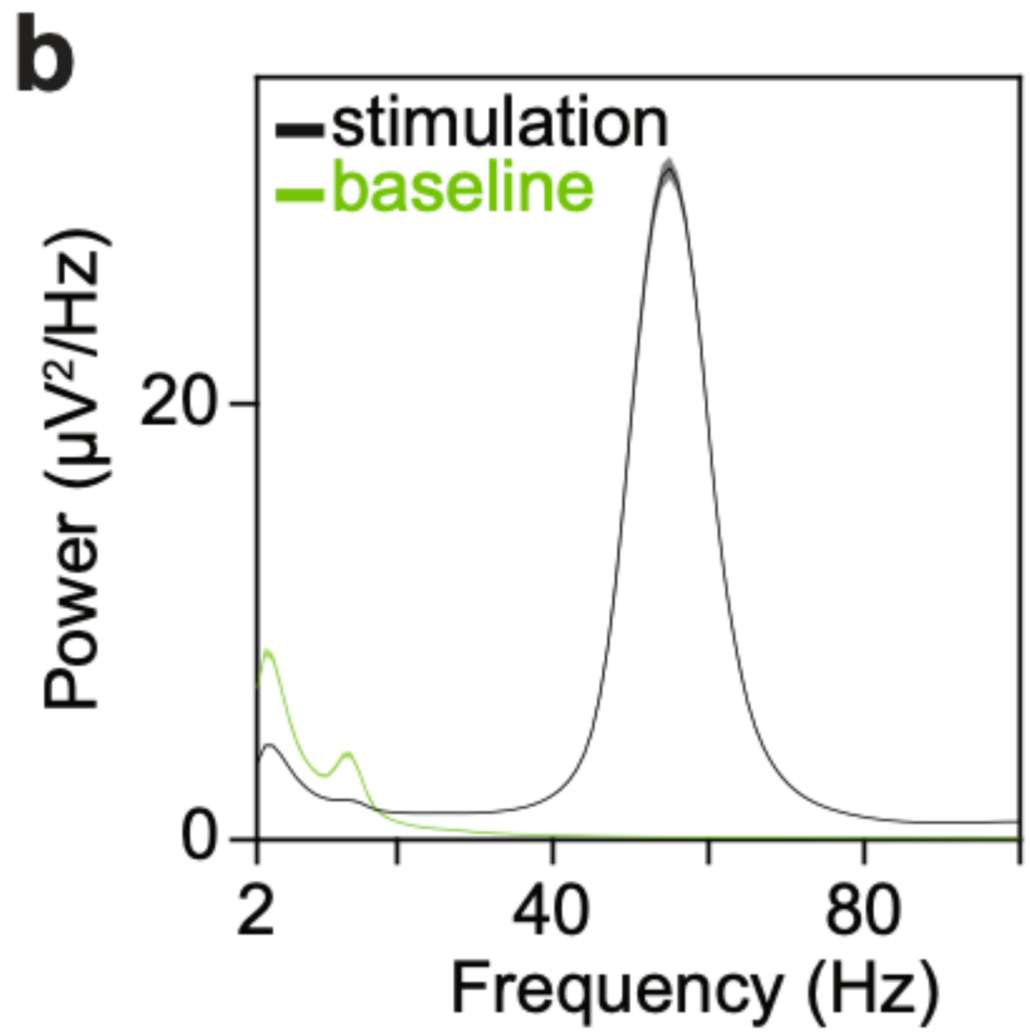
[Ainsworth et al, Neuron, 2012]

Q: What rhythms are dominant?

Example: Real world

Q: What does the power spectrum of real-world brain signals look like?

Ex. Macaque V1 during visual stimulation



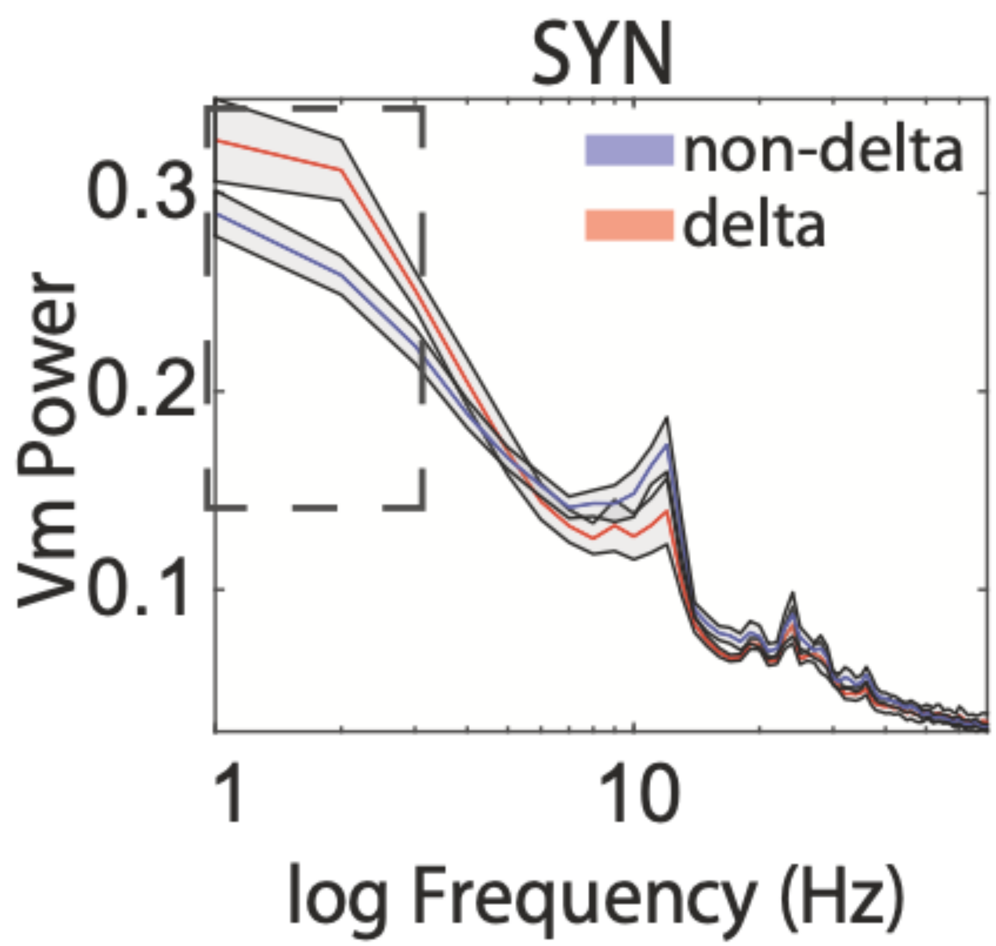
[Spyropoulos et al, Nat Comm, 2022]

Q: What rhythms are dominant?

Example: Real world

Q: What does the power spectrum of real-world brain signals look like?

Ex. Mouse striatum during locomotion.



[Shroff et al, Nat Comm, 2023]

Q: What rhythms are dominant?

Summary

Brain activity is rhythmic.

Visual inspection is powerful.

Represent data as sinusoids (sine and cosine).

The spectrum indicates the dominant sinusoids.
… the dominant rhythms.

<u>Note</u> (for the future)
- Fundamental units of AI do <u>not</u> produce rhythms.

Q. Does that matter?